

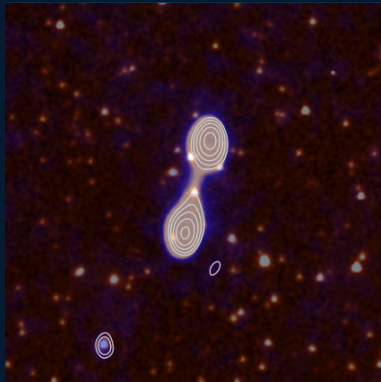
IAA-CSIC SEVERO OCHOA SCHOOL
ON AI/ML IN ASTRONOMY

Data Augmentation & Radio Classification

From FITS cutouts to pretrained CNNs
and Vision Transformers

Andrea DeMarco

Institute of Space Sciences and Astronomy,
University of Malta

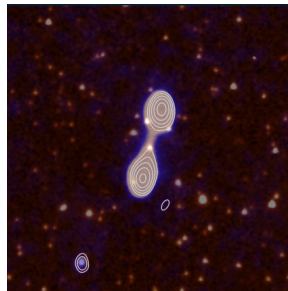


Granada | Monday 28 Sep 2026

My Research

Topics of interest

- ▶ data augmentation for radio astronomy;
- ▶ domain-adapted and foundation models for radio surveys;
- ▶ generative AI for radio morphology;
- ▶ Quantum ML applications in Astronomy;
- ▶ fast radio burst (FRB) detection;
- ▶ gravitational lensing detection.



Radio structure and its host: connecting patterns to physical objects.

How can we turn large radio-image collections into reliable scientific samples?

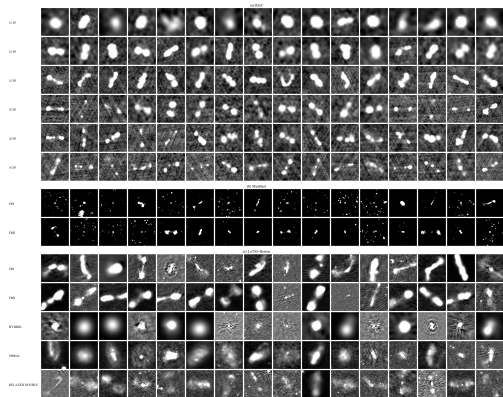
University of Malta profile; research examples follow. Image: Radio Galaxy Zoo tutorial, ATLAS radio contours over Spitzer infrared.

My observations and data

Radio interferometers

Antenna signals are correlated, calibrated and reconstructed into sky images.

- ▶ LOFAR / LoTSS;
- ▶ ASKAP survey imaging;
- ▶ MeerKAT survey imaging;
- ▶ future SKA imaging.



Resolution, noise and processing change what the same morphology looks like.

Image: our STRADAViT study, DeMarco et al. (2026).

Selected Publications: 2021-2023

CNN architectures

Magro et al., 2021

A Comparative Study of Convolutional Neural Networks for the Detection of Strong Gravitational Lensing.
Compare CNN architectures for recognising gravitational-lens candidates.

Daniel Magro, Kristian Zarb Adami, Andrea DeMarco, Simone Riggi and Eva Sciacca; doi:10.1093/mnras/stab1635.

Radio source detection

Riggi et al., 2023

Astronomical source detection in radio continuum maps with deep neural networks.
Locate and classify sources with Mask R-CNN, including extended structure and imaging artefacts.

Astronomy and Computing 42, 100682; doi:10.1016/j.ascom.2022.100682.

Detection and segmentation

Sortino et al., 2023

Radio astronomical images object detection and segmentation: A benchmark on deep learning methods.
Benchmark accuracy and computational efficiency on radio images.

Experimental Astronomy; collaboration with Renato Sortino, Daniel Magro and colleagues; doi:10.1007/s10686-023-09893-w.

Selected Publications: 2024-2026

Conditional diffusion

Sortino et al., 2024

RADiff: Controllable Diffusion Models for Radio Astronomical Maps Generation.

Generate radio images and annotations for data augmentation.

IEEE Transactions on Artificial Intelligence 5(12), 6524-6535; doi:10.1109/TAI.2024.3436538.

Unsupervised population analysis

Bufano et al., 2024

Sifting the debris: Patterns in the SNR population with unsupervised ML methods.

Convolutional autoencoders and clustering connect radio/infrared structure in supernova remnants.

Astronomy & Astrophysics 690, A203; doi:10.1051/0004-6361/202451096.

Transferable features

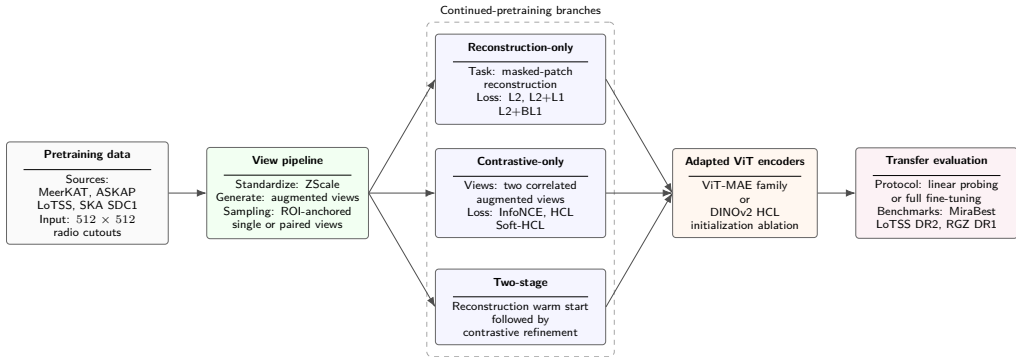
DeMarco et al., 2026

STRADAViT: Self-Supervised Domain Adaptation of Vision Transformer Backbones for Radio Astronomy.

Adapt representations to radio data before fitting a task-specific classifier.

Astronomy and Computing, in prep.; arXiv:2603.29660v4.

STRADAViT: the research framework



Mixed-survey radio images $\longrightarrow$ adapted representations $\longrightarrow$ morphology tasks.

Architecture/workflow figure reproduced from our STRADAViT manuscript. DeMarco et al. (2026), Astronomy and Computing, in prep.; arXiv:2603.29660v4.

ML4Astro: our research community

The conference series

2022 - Catania

First edition

2024 - Catania

Second edition

2026 - Valletta, Malta

31 August-4 September
University of Malta

2028 - Granada?

ML4Astro3: committees, programme, previous editions.

My involvement in ML4Astro3

- ▶ Scientific Organising Committee;
- ▶ Local Organising Committee;
- ▶ convening survey and extragalactic sessions.

A shared conversation

Astronomers and ML researchers comparing methods across surveys, time-domain astronomy, simulations and generative AI.

GRAINS: generative AI and opportunities

Generative Radio Astronomy Imaging with Neural Systems

The next research question

Can generated radio images improve classifiers when tested on real, unseen sources?

- ▶ Generate images conditioned on morphology and survey properties.
- ▶ Investigate augmentation for scarce classes.
- ▶ Test fidelity, memorisation and downstream usefulness.

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researcher positions

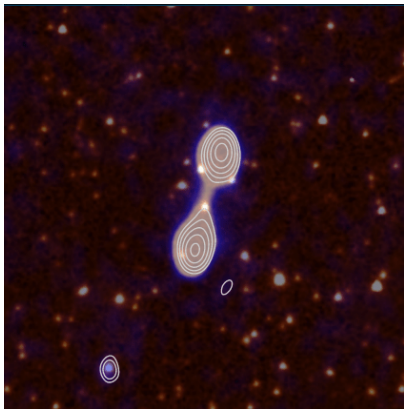
Recruiting in **2027**

24-month project

University of Malta

Interested? Contact me:
andrea.demarco@um.edu.mt

Begin as an astronomer: what is the source?



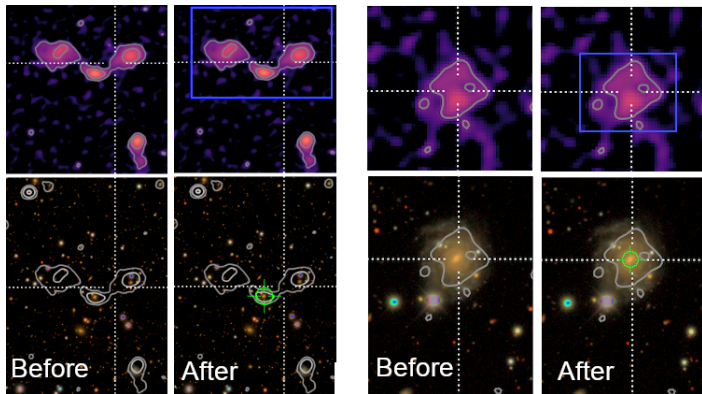
ATLAS radio contours overlaid on infrared imaging. RGZ tutorial.

Before we classify anything...

- ▶ Which radio components belong together?
- ▶ Where is the host galaxy?
- ▶ Which structure carries the morphological evidence?

Pixels do not arrive with an “object boundary” attached.

Source assembly comes before morphology



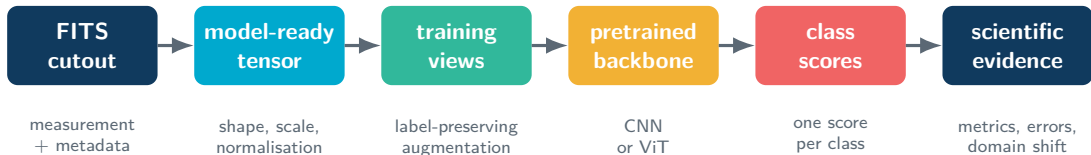
The same sky region can contain

- ▶ several radio components;
- ▶ unrelated neighbours;
- ▶ a faint or ambiguous host;
- ▶ artefacts and diffuse emission.

Classification assumes that the cutout already represents a meaningful source.

Radio Galaxy Zoo: EMU project example, Zooniverse.

The whole day follows one source



Session I: inspect data, class imbalance, scaling and augmentation.

Session II: train a transferred model, evaluate and interpret errors.

By the end, you should be able to explain four decisions

1. Input

What did the FITS cutout measure, and what enters the model?

2. Views

Which transformations preserve the morphology label?

3. Features

How do a CNN and a ViT represent a radio source?

4. Evidence

Which results justify trusting - or rejecting - the model?

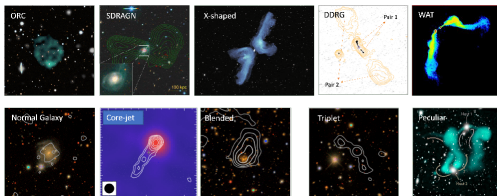
You do not need to memorise architectures; you need to reason about the pipeline.

01 FROM SKY TO DATASET

What exactly enters the classifier?

Build the physical picture before the model.

Radio morphology is richer than two tidy boxes



A morphology label compresses several questions at once

Components

compact, double, triple

Geometry

straight, bent, X-shaped

Brightness pattern

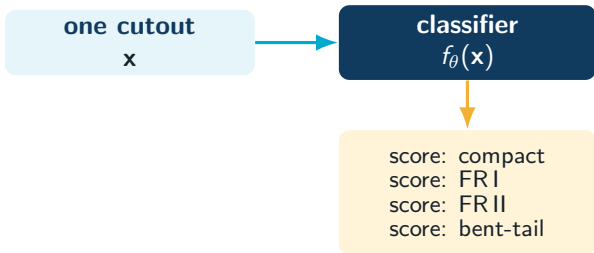
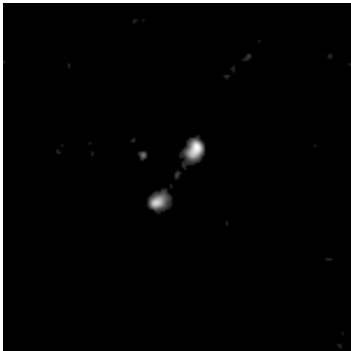
hotspots, diffuse lobes

History / context

restarted, blended, peculiar

Nature overlaps these axes; the workshop uses a small, declared class vocabulary to make one testable task.

What is a multi-class classification task?



- ▶ The model returns **one score for every allowed class**.
- ▶ The highest score becomes the prediction.
- ▶ The class list is part of the scientific question.

A model cannot discover a class that the target definition does not permit.

Labels are useful measurements - not perfect truth

Multi-class

one label from a list

FR I *or* FR II

Multi-label

several properties

bent + restarted + diffuse

Uncertain

disagreement or low
confidence

do not hide ambiguity

- ▶ Catalogues encode selection rules, expert judgement and survey limitations.
- ▶ A clean training target can still be an imperfect description of nature.

Workshop contract: predict one declared class, but keep disagreement or low confidence visible for review.

A FITS cutout is data plus a coordinate system

HEADER

```
NAXIS1 = 150  
NAXIS2 = 150  
CDELTA1 =  $-5.0 \times 10^{-4}$  deg  
CDELTA2 =  $5.0 \times 10^{-4}$  deg  
BMAJ =  $1.5 \times 10^{-3}$  deg  
BMIN =  $1.2 \times 10^{-3}$  deg  
BUNIT = 'Jy/beam'
```

The array tells us *values*.
The header tells us *meaning*.

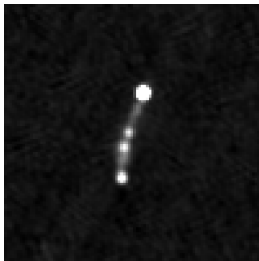
- ▶ NAXIS1/2: image dimensions;
- ▶ CDELTA1/2: pixel scale and axis direction;
- ▶ BMAJ/BMIN: synthesised beam axes;
- ▶ BUNIT: flux-density unit.

Two 150×150 arrays need not represent the same angular or physical scale.

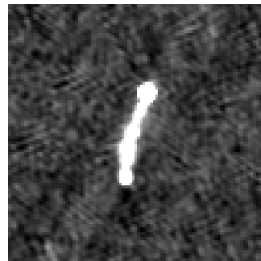
One FITS array, two different jobs



Measured FITS values
linear display over full min-max



Human inspection
one option: z-scale display limits



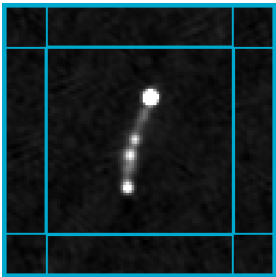
Model input
background/noise-scaled tensor

Z-scale is one display option, not a universal prescription. A model needs an explicit preprocessing rule.

Compare native, percentile and z-scale views. A stretch used as model input also changes the model's numbers.

Real Radio Galaxy Zoo DR1 FITS cutout; display and model routes shown separately.

Background and noise define one useful model scale



Choose a background region
no detectable source emission

Off-source means away from detectable source emission.
 I_{off} is the set of pixel values in that chosen region.
Estimate the background level b and noise scatter σ from it.

$$b = \text{median}(I_{\text{off}}) \quad \sigma = 1.4826 \text{ median } |I_{\text{off}} - b|$$

subtract

$$I - b$$



scale

$$(I - b)/\sigma$$



bound

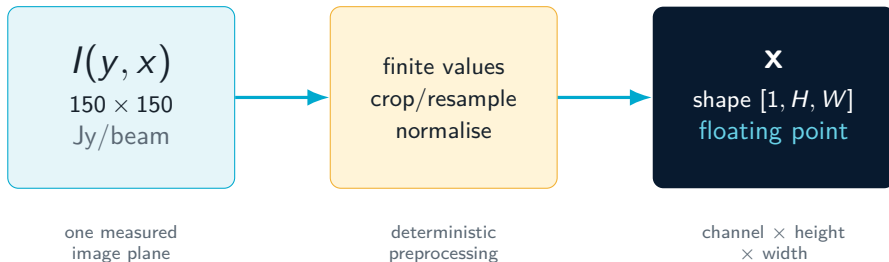
clip, e.g.
 $[-3, 10]$

The result is in approximate noise units. Check that the border excludes faint lobes, other sources and imaging artefacts.

Fit preprocessing on training data or a declared off-source rule; apply it unchanged to validation, test and deployment.

Illustrative background border on the same RGZ DR1 cutout. The median absolute deviation estimates noise robustly; masks and correlated noise need care.

From one FITS plane to one model tensor



Record every operation that changes the numbers, shape or physical interpretation.

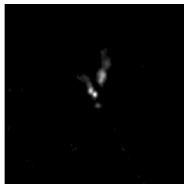
A dataset contains variation within every class



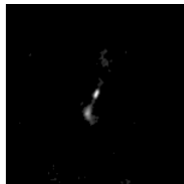
FR I



FR I



FR I



FR I



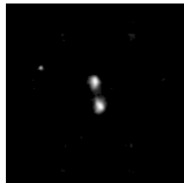
FR II



FR II



FR II



FR II

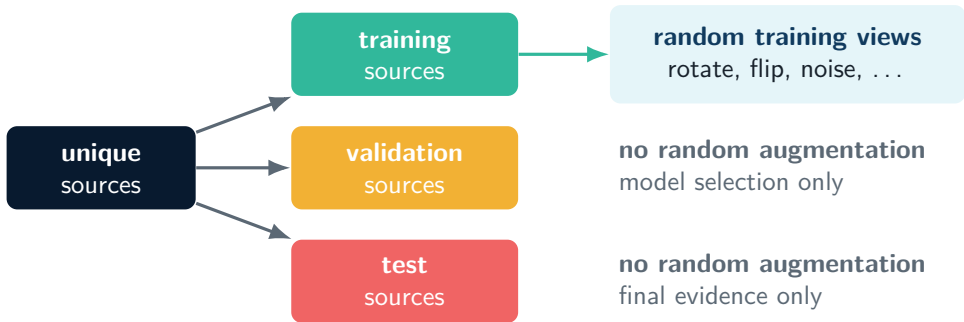
orientation
arbitrary on the sky

angular extent
compact to extended

surface brightness
bright to diffuse

Genuine MiraBest cutouts from the public Zenodo archive; display-only conservative percentile limits; original arrays unchanged.

Split sources before the model sees transformed copies



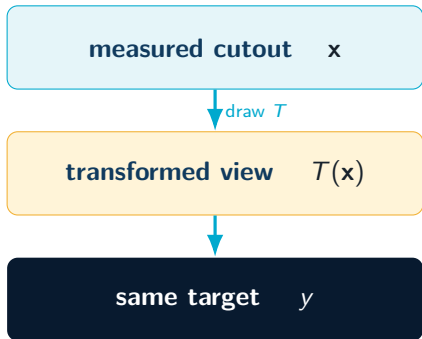
If two views of the same source cross the split, the test is no longer independent.

02 DATA AUGMENTATION

Different view, same answer

Build the physical picture before the model.

Augmentation teaches what should not change the answer



The central question

Would an astronomer assign the same class after this change?

If yes, the transformed view teaches the model to ignore that nuisance. The technical name is an **invariance**.

Augmentation is a scientific claim about label preservation.

One source, several possible training views



USUALLY SAFE
orientation

TEST THE ASSUMPTION
observing conditions

LIKELY UNSAFE
evidence removed

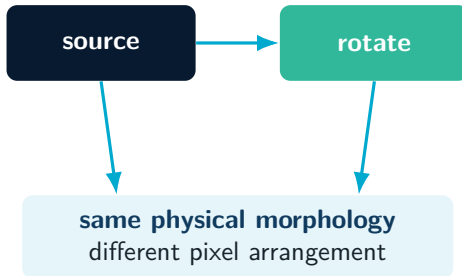
Transforms shown on a genuine MiraBest FR II cutout; greyscale percentile limits fixed from the original cutout across all views.

Rotation and reflection: the clean starting point

Physical statement

A radio galaxy's sky orientation should not determine its morphology class.

- ▶ 90° rotations and flips reorder pixels exactly.
- ▶ Arbitrary angles require interpolation.
- ▶ Interpolation also changes noise correlations.



Start with the transformation whose physical justification is clearest.

Noise, beam and crop change the observing problem

Noise

Can simulate a depth range,
but may erase faint jets.

Beam blur

Can simulate resolution, but
can merge core and lobes.

Crop / shift

Can teach centring tolerance,
but may remove a lobe.

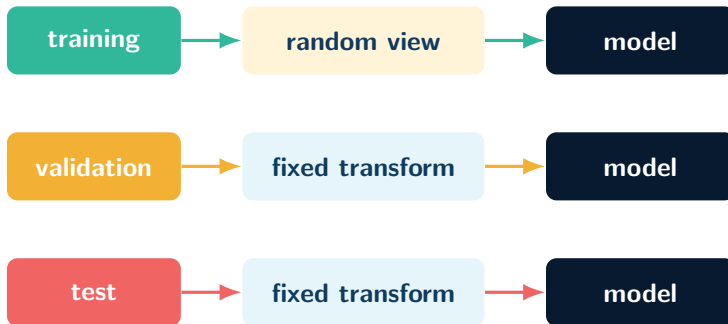
deployment survey

realistic parameter range

augmentation

Choose the transform strength from a real observing domain, not convenience.

Random augmentation belongs only in the training path



Validation compares models; the test set supports the final claim. Keep both stable.

What augmentation can - and cannot - provide

It can help the model

- ▶ ignore arbitrary orientation;
- ▶ tolerate a specified noise/beam range;
- ▶ reduce overfitting to pixel arrangements.

It cannot create

- ▶ new independent astrophysical sources;
- ▶ missing morphologies;
- ▶ unbiased labels;
- ▶ proof of cross-survey reliability.

A transformed observation is still derived from the same observation.

Quick decision: does the class survive?

A

rotate 90°

B

double the noise

C

blur to a larger beam

D

crop one lobe

Safe, conditional, or unsafe?

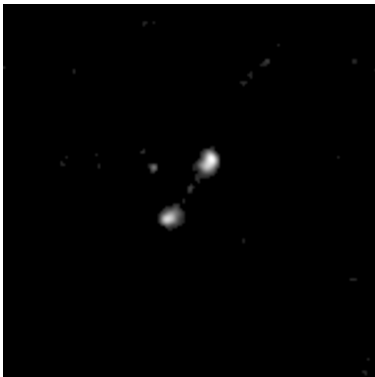
State the observing assumption behind your answer.

03 CONVOLUTIONAL NETWORKS

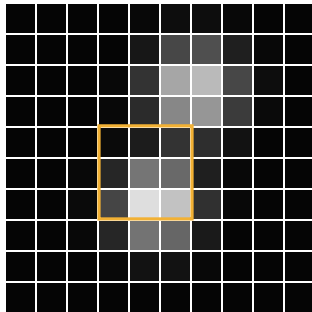
From pixels to morphology, one local pattern at a time

Build the physical picture before the model.

A neural network begins with numbers, not lobes



Our interpretation
core, jet, lobe, hotspot



The input
a grid of intensity values

The network must build useful patterns from simple numerical operations.

A filter asks one small question

-1	0	1
-1	0	1
-1	0	1

Example initial question: vertical edge?

- ▶ Place the small weight grid on an image patch.
- ▶ Multiply matching entries.
- ▶ Add the results.
- ▶ A large response means: “this pattern is present here.”

During training, the 3×3 numbers are **learned**; “edge detector” is only an interpretable starting example.

The filter is the shared weight grid; its value at every location forms one feature map.

One output pixel: multiply, then add

A small numerical example

1	2	3
4	5	6
7	8	9

 $K = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$

same 2×2 weights
at every position

input: 3×3

Start at the highlighted patch

$$1(1) + 2(0) + 4(0) + 5(-1) = \boxed{-4}$$

Move one pixel right: $2 - 6 = -4$.

Repeat on the next row: $4 - 8 = -4$, $5 - 9 = -4$.

$$\text{response map} = \begin{bmatrix} -4 & -4 \\ -4 & -4 \end{bmatrix}$$

This filter measures a local difference, not the brightness itself. A negative response is valid.

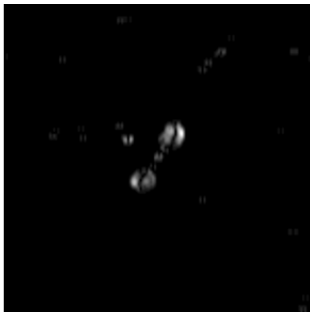
A convolutional layer repeats a local weighted sum; training learns which weights are useful.

Adapted from Andrea DeMarco, PHY3287 Lecture 6. Toy numbers; stride 1, no padding, bias zero, before activation. CNN libraries use cross-correlation (no kernel flip).

Sliding the filter creates a response map



input cutout



vertical-edge response

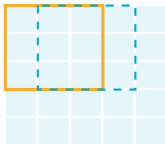
1. apply the same 3×3 filter everywhere;
2. record one response at every position;
3. bright responses mark matching local structure.

Weight sharing lets one detector recognise a pattern wherever it appears.

Illustration on a genuine MiraBest cutout; fixed vertical-edge filter.

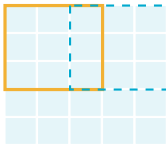
Stride and padding control where the filter looks

stride 1, no padding



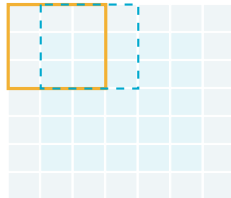
output: 3×3

stride 2, no padding



output: 2×2

stride 1, padding 1



output: 5×5

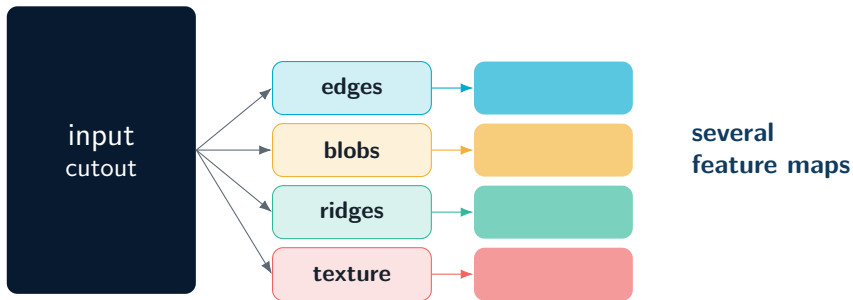
Stride = step size. A larger step reduces spatial sampling and computation; small features can be missed.

Padding = an added border. Here, one ring of zeros preserves the size at stride 1; it is not new sky data.

All three use the same 5×5 image and 3×3 kernel; only the placement rules change.

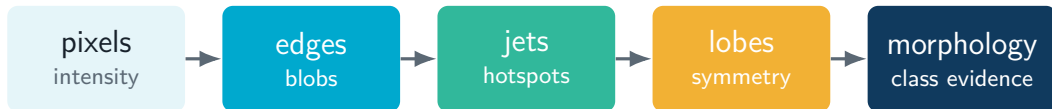
Adapted from Andrea DeMarco, PHY3287 Lecture 6. Gold: first placement; cyan dashed: next placement. Grey border: zero padding.

A convolutional layer learns many filters



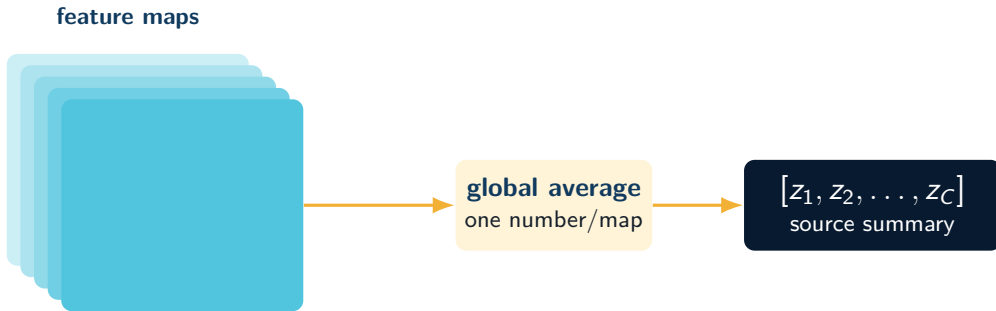
The number of output channels is the number of learned feature maps.

Depth builds a hierarchy of visual evidence



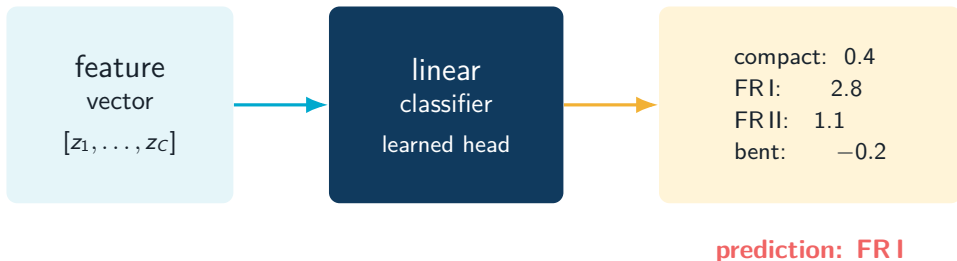
Early features are simple. Later features combine larger parts of the source.

Pooling turns feature maps into a compact summary



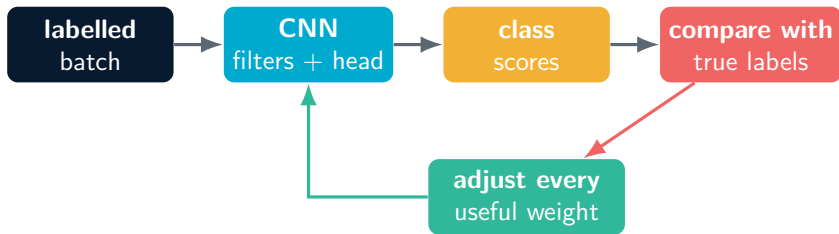
The summary says which learned patterns are present across the source.

The classifier head converts features into class scores



Raw class scores are called logits; softmax converts them into relative probabilities.

How do the filters become useful?



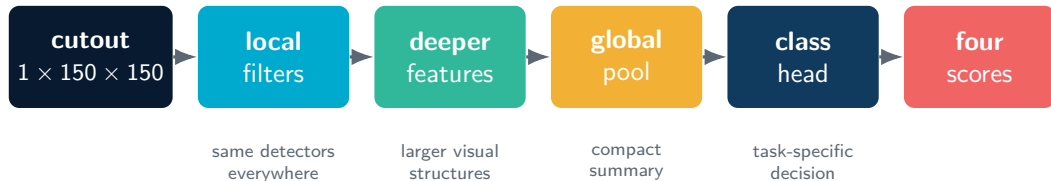
Forward pass
make predictions

Loss
measure the mismatch

Backpropagation
assign useful changes

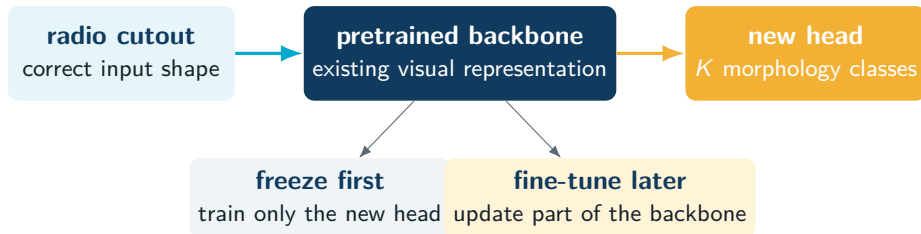
Training repeatedly changes filters and the head so that labelled examples receive better scores.

A CNN, end to end



CNN built-in assumption: nearby pixels are related, and useful patterns repeat across the image.

Pretraining gives the backbone a useful starting point



Why it may transfer: early shape and texture features can help. **Why it may fail:** RGB statistics, natural-image texture and radio noise/beam structure differ.

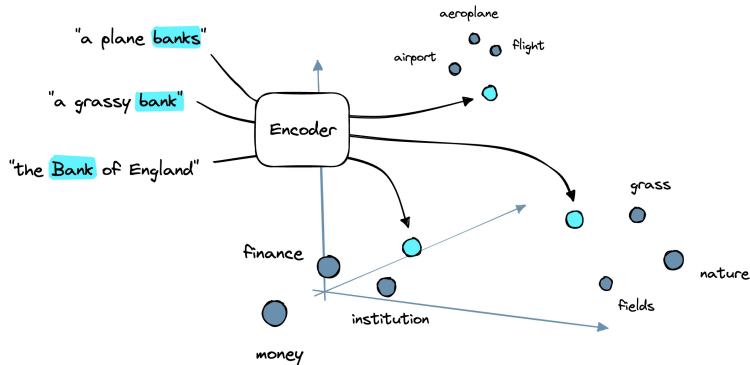
Honour the input contract, train the new head first, then test whether careful fine-tuning improves held-out radio sources.

04 FROM TRANSFORMERS TO VISION

Understand a part by looking at its context

Build the physical picture before the model.

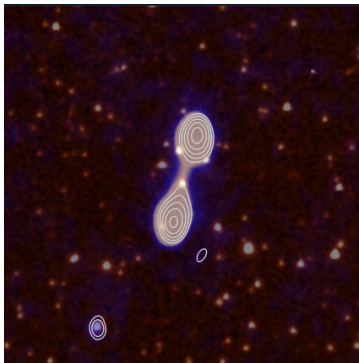
Transformer Encoder Intuition



An encoder builds a description of each word in context, not just of the word in isolation.

Andrea DeMarco, PHY3287 Lecture 8: original encoder_intuition.png. Schematic feature space, not measured embeddings.

From words to images: a bright patch also needs context



Which structures belong together?

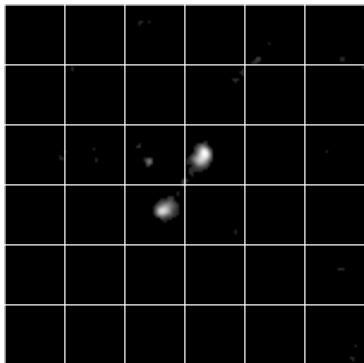
A bright patch on its own is ambiguous

- ▶ Is it a core, a hotspot, or an unrelated source?
- ▶ Does emission connect it to another component?
- ▶ Is there a corresponding lobe on the other side?

In language, words provide context. In a Vision Transformer, **image patches** provide context for one another.

The useful idea is the same: interpret each part using information from the rest.

Step 1: cut the image into a regular patch grid



The grid is imposed on the cutout
it does not follow source boundaries

Illustrative example

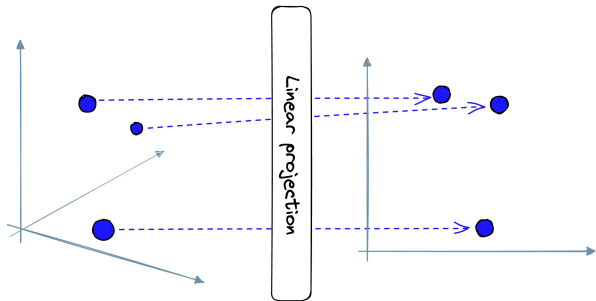
- ▶ image: 150×150 pixels;
- ▶ patch: 25×25 pixels;
- ▶ grid: 6×6 ;
- ▶ sequence: **36 patch items**.

Each patch is flattened and projected to a feature vector.

Patch size trades fine detail against sequence length and compute.

Patch grid shown on a genuine MiraBest cutout; dimensions are pedagogical.

Step 2: each patch becomes a compact descriptor



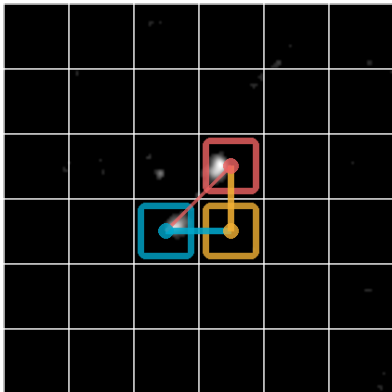
Each dot stands for one patch's numerical description.
The arrows show the learned conversion, not motion across the sky.

1. **Flatten:** turn each patch into a list of pixel values.
2. **Describe:** a learned mapping turns that list into a short numerical description of the patch.
3. **Locate:** include its grid position, so the model can distinguish where patches came from.

A descriptor is also called a **token embedding**.
It is not a morphology label.

A patch token carries visual content plus position, ready to exchange information with other tokens.

Step 3: attention asks which other patches matter



For one patch, ask

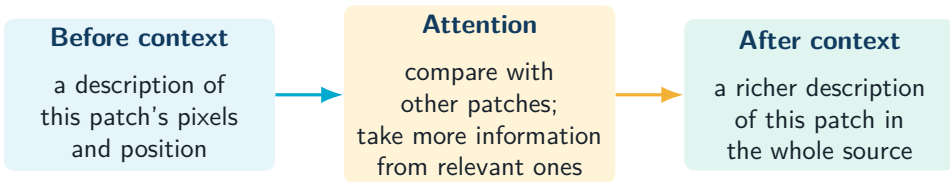
- ▶ Which other patches are relevant?
- ▶ How strongly should each contribute?
- ▶ What information should be combined?

The weights depend on the current image content.

Each patch borrows information from relevant patches; how much it borrows depends on this image.

Illustrative patch relations on a genuine MiraBest cutout - not measured attention weights.

What changes when patches exchange information?



Not just the nearest neighbour.

A distant patch can contribute strongly if its features are relevant. Several patches can contribute at once.

Not a change to the image.

The pixel values stay the same. It is the model's internal description that gains context.

An encoder repeatedly exchanges information, then refines each description.

Adapted from Andrea DeMarco, PHY3287 Lectures 7-8. Attention is one part of an encoder block.

Several attention heads provide several learned views

Head A

one learned pattern of
patch-to-patch weights

Head B

a different learned pattern of
weights

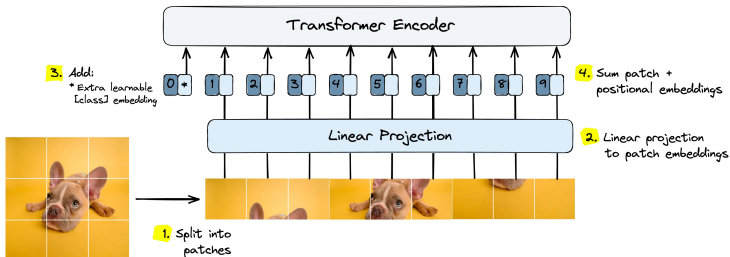
Head C

another learned pattern of
weights

The outputs are combined. A head does not arrive with a human-readable physical meaning.

Attention weights show information routing - not automatically causal explanation.

The [CLS] token: one summary for the whole image



1. Add a summary token

[CLS] is an extra learned input, not an image patch or a known label.

2. Gather context

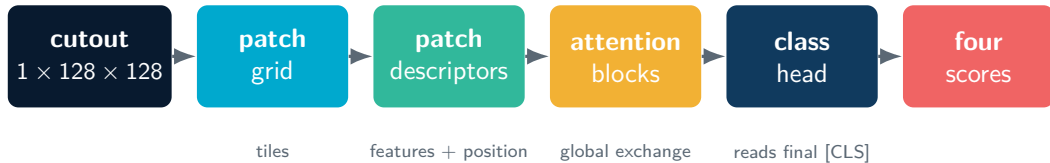
It exchanges information with patch tokens through the encoder.

3. Classify the image

Its final description goes to the classifier head to produce class scores.

Many patch descriptions become one image-level decision.

A Vision Transformer, end to end



ViT built-in assumption: positioned image regions can form useful learned relationships.

CNN and ViT: two routes to a source representation

CNN

- ▶ starts with small local filters;
- ▶ shares them across locations;
- ▶ builds global structure through depth;
- ▶ strong, data-efficient image bias.

ViT

- ▶ starts with patch descriptors;
- ▶ connects patches through attention;
- ▶ expresses long-range relations directly;
- ▶ usually benefits strongly from pretraining.

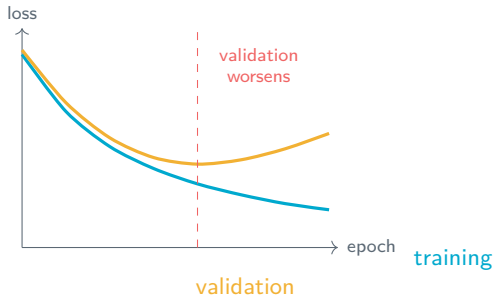
Compare architecture under the same sources, labels, preprocessing, augmentation, tuning budget and metrics.

05 TRAINING & EVIDENCE

A prediction is not yet a scientific result

Build the physical picture before the model.

Learning curves tell a story



Three words

- ▶ **batch**: a subset used for one weight update;
- ▶ **epoch**: one pass through the training sources;
- ▶ **validation**: separate sources used to check progress, not update weights.

Is training improving? Does validation improve too?
When does the gap grow?

Save the model at the chosen validation point - not simply the final epoch.

A confusion matrix shows *which* mistakes occur

		predicted class			
		compact	FR I	FR II	bent
true class	compact	64	5	3	8
	FR I	7	70	19	4
	FR II	3	11	76	10
	bent	8	9	5	82

- Read **across a true-class row**.
- Diagonal cells are correct examples.
- Off-diagonal cells are specific mistakes.
- Highlighted: **19 true FR I sources were predicted FR II**.
- Recall: how much of a true class is recovered.

For imbalanced classes, report **per-class** and **macro-averaged** metrics.

Inspect the images behind the errors



CASE A



CASE B



CASE C

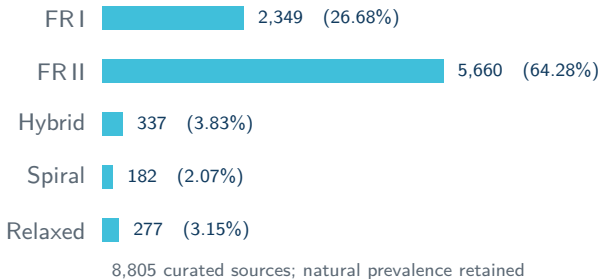
- ▶ **Confident error:** shortcut, wrong label, or missing class?
- ▶ **Uncertain case:** overlap, low resolution, or low signal-to-noise?
- ▶ **Model disagreement:** do CNN and ViT use different evidence?

For each case, show the image, true label, prediction, confidence and relevant metadata.

Failure analysis connects a metric back to morphology, data quality and label design.

Example review images from the public MiraBest archive; not model outputs.

A real LoTSS cohort is strongly imbalanced



64.3% accuracy

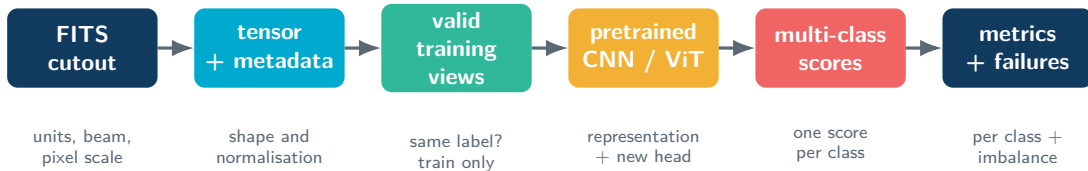
comes from always predicting FR II.

- ▶ **Weighted loss:** changes the cost of each error.
- ▶ **Weighted sampler:** changes which training examples appear.
- ▶ **Evaluation:** keep validation and test prevalence natural.

Report macro F1, balanced accuracy and every class recall - not accuracy alone.

Horton et al. (2025), LoTSS DR2 visual-classification catalogue; unambiguous five-class subset.

The complete reasoning chain



Every arrow is an assumption that can be documented and tested.

What the hands-on sessions will ask you to do

Radio Survey Pipelines I

1. lazily load and inspect real FITS;
2. compare scaling; choose and justify a suitable stretch;
3. quantify imbalance and metadata;
4. defend augmentation; export the data contract.

Radio Survey Pipelines II

1. load a pretrained CNN or ViT;
2. compare loss/sampling strategies;
3. select on validation; test once;
4. inspect per-class errors and defend a choice.

Session I: freeze the data contract → **Session II:** compare controlled model evidence.

The notebook scaffolds the pipeline; you investigate and defend the choices.

TAKE-HOME MODEL

The classifier is only one component of the scientific pipeline.

Understand the measurement. Preserve the label.

Compare representations fairly. Inspect the failures.

Next: Radio Survey Pipelines I & II

BACKUP

Equations and extensions

Build the physical picture before the model.

Plain-language glossary

- ▶ **tensor**: a numerical array with a declared shape;
- ▶ **channel**: one image plane, such as radio intensity;
- ▶ **filter**: a small grid of learned CNN weights;
- ▶ **feature map**: the filter response at every location;
- ▶ **patch descriptor**: a learned feature vector for one image patch.
- ▶ **backbone**: the main feature extractor;
- ▶ **head**: the task-specific output layers;
- ▶ **logit**: one raw class score;
- ▶ **batch**: examples used for one update;
- ▶ **epoch**: one pass through the training set.

Training changes weights. Validation selects choices. Testing supports the final claim.

Mathematical Definition of Convolution

- ▶ For a discrete 2D image f and kernel g , the convolution at output pixel (i, j) (in the CNN sense, actually cross-correlation) is:

$$(f * g)[i, j] = \sum_u \sum_v f[i+u, j+v] \cdot g[u, v].$$

- ▶ Here, (i, j) index the *output* location; (u, v) range over the kernel support.
 - $f[i+u, j+v]$: the input pixel under the kernel position (u, v) , measured relative to the current top-left placement of the kernel on the image.
 - $g[u, v]$: the kernel *weight* at offset (u, v) ; it scales that input pixel.
 - Sum the products over all (u, v) in the kernel to produce the output value at (i, j) .
- ▶ This operation slides the kernel across the image, computing weighted sums of local neighborhoods.

Copied from Andrea DeMarco, PHY3287 Lecture 6.

Step-by-Step Convolution Example (1/2)

- ▶ Input matrix (3x3):

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

- ▶ Kernel (2x2):

$$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

- ▶ Slide kernel over input, compute element-wise product sum for each position:

- **Top-left:** overlay kernel on input positions (1, 2, 4, 5):

$$(1 * 1) + (2 * 0) + (4 * 0) + (5 * (-1)) = 1 + 0 + 0 - 5 = -4$$

- **Top-right:** overlay kernel on input positions (2, 3, 5, 6):

$$(2 * 1) + (3 * 0) + (5 * 0) + (6 * (-1)) = 2 + 0 + 0 - 6 = -4$$

Step-by-Step Convolution Example (2/2)

- ▶ **Bottom-left:** overlay kernel on input positions (4, 5, 7, 8):

$$(4 * 1) + (5 * 0) + (7 * 0) + (8 * (-1)) = 4 + 0 + 0 - 8 = -4$$

- ▶ **Bottom-right:** overlay kernel on input positions (5, 6, 8, 9):

$$(5 * 1) + (6 * 0) + (8 * 0) + (9 * (-1)) = 5 + 0 + 0 - 9 = -4$$

- ▶ The resulting 2×2 output matrix is:

$$\begin{bmatrix} -4 & -4 \\ -4 & -4 \end{bmatrix}$$

Copied from Andrea DeMarco, PHY3287 Lecture 6.

Queries, Keys, Values (Q/K/V) - Intuition

- ▶ We now formalize the spotlight idea with Q (what to look for), K (where it is), and V (what to take).
- ▶ **Query (Q)**: what this position seeks
- ▶ **Key (K)**: what each position offers
- ▶ **Value (V)**: the content to aggregate
- ▶ $\text{Similarity}(Q_i, K_j) \Rightarrow$ how much V_j contributes to output at i

Copied from Andrea DeMarco, PHY3287 Lecture 7.

Shapes and Book-keeping (1/2)

► **Input tokens:** $X \in \mathbb{R}^{T \times d}$ (sequence length T , feature dim d).

- Example (text): if a sentence has 12 tokens, $T = 12$.
- Example (image): if a 224×224 image is split into 16×16 patches, $T = 14 \times 14 = 196$.

► **Linear projections:**

$$Q = XW_Q, \quad K = XW_K, \quad V = XW_V,$$

with $W_Q, W_K, W_V \in \mathbb{R}^{d \times d_k}$ so $Q, K, V \in \mathbb{R}^{T \times d_k}$.

- *Why projections?* They let the model learn *separate roles*: a set of learned subspaces.
- W_Q, W_K, W_V are **learned parameter matrices** ($d \times d_k$). Multiplying X by them linearly mixes features to form queries/keys/values.
- Later on, in multi-head attention, each head will have its own (W_Q^h, W_K^h, W_V^h)!
- *Capacity/control*: Decoupling Q/K (scoring space) from V (content space) increases expressivity; choosing d_k controls per-head dimensionality (often $d_k = d/h$).

Copied from Andrea DeMarco, PHY3287 Lecture 7.

Shapes and Book-keeping (2/2)

- ▶ **Similarity scores (all pairs):**

$$S = \frac{QK^T}{\sqrt{d_k}} \in \mathbb{R}^{T \times T}.$$

Row i are scores of query i against every key j .

- ▶ **Attention weights (row-wise softmax):**

$$A = \text{softmax}(S) \in \mathbb{R}^{T \times T}, \quad \sum_j A_{ij} = 1.$$

- ▶ **Weighted mixing of values:**

$$Y = AV \in \mathbb{R}^{T \times d_k}.$$

Row i in Y is the context-enhanced representation of token i .

Copied from Andrea DeMarco, PHY3287 Lecture 7.

Why Softmax?

- ▶ Converts arbitrary scores to a **probability-like** distribution over sources.
- ▶ Emphasizes the top matches, but still allows mixing if several are relevant.
- ▶ Temperature (not used here) would control how *peaky* the focus is.

Suppose a query has dot-products $[2, 1, 0]$ with three keys. Softmax gives $\alpha \approx [0.665, 0.245, 0.090]$.
Weighted sum $\sum \alpha_i v_i$ copies more from the first value but still blends information.

Copied from Andrea DeMarco, PHY3287 Lecture 7.

Worked Toy Example (3 tokens)

Let $d = 2$ and $Q, K, V \in \mathbb{R}^{3 \times 2}$ (three tokens, $\dim = 2$):

$$Q = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}, \quad K = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}, \quad V = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}.$$

Scores: $S = \frac{QK^\top}{\sqrt{2}} = \begin{bmatrix} \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ 0 & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & \frac{2}{\sqrt{2}} \end{bmatrix} \approx \begin{bmatrix} 0.707 & 0 & 0.707 \\ 0 & 0.707 & 0.707 \\ 0.707 & 0.707 & 1.414 \end{bmatrix}.$

Row-wise softmax: $A = \text{softmax}(S) \approx \begin{bmatrix} 0.401 & 0.198 & 0.401 \\ 0.198 & 0.401 & 0.401 \\ 0.248 & 0.248 & 0.504 \end{bmatrix}, \quad \sum_j A_{ij} = 1 \ \forall i.$

Output: $Y = AV \approx \begin{bmatrix} 0.802 & 0.599 \\ 0.599 & 0.802 \\ 0.752 & 0.752 \end{bmatrix}.$

Each row of A is the attention distribution for that token; Y_i is the *context-enriched* representation of token i .

Copied from Andrea DeMarco, PHY3287 Lecture 7.

High-Level ViT Idea

- ▶ Split an image into non-overlapping patches (e.g., $P = 16 \Rightarrow 14 \times 14 = 196$ patches for 224×224).
- ▶ Flatten each patch $\mathbb{R}^{P \times P \times C} \rightarrow \mathbb{R}^{P^2 C}$ and linearly project to dim d via $E \in \mathbb{R}^{P^2 C \times d}$.
- ▶ Prepend a learnable [CLS] token; add positional encodings; send to Transformer encoder blocks (Dosovitskiy et al., 2021).
- ▶ Final [CLS] = global summary; a linear classifier head turns it into logits.
- ▶ *Example:* $224 \times 224 \times 3 \rightarrow 14 \times 14 = 196$ patches, sequence length 197 including [CLS].

Copied from Andrea DeMarco, PHY3287 Lecture 8.

Metrics in Astronomy

- ▶ **Accuracy** misleading if classes imbalanced
- ▶ **Precision**: fraction of predicted positives that are true
- ▶ **Recall**: fraction of actual positives recovered
- ▶ **F1-score**: harmonic mean of precision and recall
- ▶ In astronomy: often prioritize recall (completeness) or precision (purity)

Copied from Andrea DeMarco, PHY3287 Lecture 4.

F1-Score

- ▶ $F1 = 2 * (\text{precision} * \text{recall}) / (\text{precision} + \text{recall})$
- ▶ F1 score can be interpreted as a harmonic mean of the precision and recall.
- ▶ F1 score reaches its best value at 1, and worst at 0

Copied from Andrea DeMarco, PHY3287 Lecture 4.

Validating synthetic augmentation

1. Check train/test contamination and memorisation.
2. Test morphology fidelity with domain experts.
3. Measure diversity, not only visual quality.
4. Compare real-only and real+synthetic training on one frozen real test set.
5. Inspect whether the classifier learns generator fingerprints.

Synthetic samples are model outputs - not independent astronomical evidence.

Selected reading

- ▶ Fanaroff & Riley (1974), the original FR morphology division.
- ▶ Porter & Scaife (2023), MiraBest: a morphologically classified radio-galaxy dataset.
- ▶ Horton et al. (2025), complex morphology in LoTSS DR2.
- ▶ Lukic et al. (2018), Radio Galaxy Zoo classification with deep learning.
- ▶ Scaife & Porter (2021), group-equivariant CNNs for FR classification.
- ▶ Dosovitskiy et al. (2021), An Image Is Worth 16×16 Words.
- ▶ Slijepcevic et al. (2023), self-supervised representations for radio astronomy.

Image and data provenance

- ▶ MiraBest article and data description:
doi.org/10.1093/rasti/rzad017
- ▶ MiraBest public data archive used for the cutout figures:
zenodo.org/records/4288837
- ▶ Radio Galaxy Zoo source-assembly tutorial:
Galaxy Zoo blog: a close-up look at one example galaxy
- ▶ Radio Galaxy Zoo: EMU research and morphology guide:
Zooniverse: Radio Galaxy Zoo EMU
- ▶ Scaling demonstration: bundled RGZ DR1 FITS cutout `RGZ_J120115.4+180947....fits`; display and tensor figures are derived locally.

Preserve the measured arrays. Choose display limits for the dataset, and document model preprocessing.